

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{144} \frac{1}{m_c^2} \left(c_\gamma^2 \left(-g_2^2 \overline{y}_d^{12p} y_u^{11p} - 12 s_\gamma^2 \left(\overline{y}_d^{rp} \overline{y}_d^{12s} y_d^{rs} y_d^{11p} + \overline{y}_u^{12s} \left(3 \overline{y}_d^{pr} y_d^{11r} y_u^{ps} + \right. \right. \right. \right. \right. \\
& \quad \left. \left. \left. \left. \overline{y}_u^{rp} y_u^{rs} y_u^{11p} \right) \right) \right) - s_\gamma^2 \overline{y}_d^{12p} \left(g_2^2 y_d^{11p} + 36 c_\gamma^2 y_d^{rp} \overline{y}_u^{rs} y_u^{11s} \right) \right) + \\
& \frac{1}{18} g_2^4 \text{LF}_{3,0}[\text{m}_2] \delta_{11i2} + \frac{1}{6} g_2^4 \text{LF}_{4,-1}[\text{m}_2] \delta_{11i2} - \frac{8}{45} g_2^4 \text{LF}_{5,-2}[\text{m}_2] \delta_{11i2} + \\
& \frac{1}{18} \sum p g_2^4 \text{LF}_{3,0}[\text{m}_1^p] \delta_{11i2} - \frac{5}{48} \sum p g_2^4 \text{LF}_{4,-1}[\text{m}_1^p] \delta_{11i2} + \\
& \frac{1}{24} \sum p g_2^4 \text{LF}_{5,-2}[\text{m}_1^p] \delta_{11i2} + \frac{1}{6} \sum p g_2^4 \text{LF}_{3,0}[\text{m}_q^p] \delta_{11i2} - \\
& \frac{1}{15} \sum p g_2^4 \text{LF}_{4,-1}[\text{m}_q^p] \delta_{11i2} + \frac{2}{15} \sum p g_2^4 \text{LF}_{5,-2}[\text{m}_q^p] \delta_{11i2} - \\
& \frac{1}{24} g_2^2 \left(s_\gamma^2 \overline{y}_d^{12p} y_d^{11p} + c_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \right) \text{LF}_{2,1}[\text{m}_0] + \\
& \frac{1}{72} g_2^2 \left(3 s_\gamma^2 \overline{y}_d^{12p} y_d^{11p} + 3 c_\gamma^2 \overline{y}_u^{12p} y_u^{11p} + 4 g_2^2 \delta_{11i2} \right) \text{LF}_{3,0}[\text{m}_0] - \\
& \frac{5}{48} g_2^4 \text{LF}_{4,-1}[\text{m}_0] \delta_{11i2} + \frac{2}{45} g_2^4 \text{LF}_{5,-2}[\text{m}_0] \delta_{11i2} + \frac{1}{18} g_2^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{11i2} + \\
& \frac{1}{12} g_2^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{11i2} - \frac{4}{45} g_2^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{11i2} + \frac{1}{18} g_1^2 c_\gamma^2 \overline{y}_d^{12p} y_d^{11p} \text{LF}_{2,1,0}[\text{m}_1, \text{m}_d^p] - \\
& \frac{1}{9} g_1^2 c_\gamma^2 \overline{y}_d^{12p} y_d^{11p} \text{LF}_{3,1,-1}[\text{m}_1, \text{m}_d^p] + \frac{1}{18} g_1^2 c_\gamma^2 \overline{y}_d^{12p} y_d^{11p} \text{LF}_{4,1,-2}[\text{m}_1, \text{m}_d^p] + \\
& \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,1,0}[\text{m}_1, \text{m}_q^{11}] \delta_{11i2} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,2,-1}[\text{m}_1, \text{m}_q^{11}] \delta_{11i2} - \\
& \frac{1}{432} g_1^2 g_2^2 \text{LF}_{3,1,-1}[\text{m}_1, \text{m}_q^{11}] \delta_{11i2} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{4,1,-2}[\text{m}_1, \text{m}_q^{11}] \delta_{11i2} + \\
& \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,1,0}[\text{m}_1, \text{m}_q^{12}] \delta_{11i2} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{2,2,-1}[\text{m}_1, \text{m}_q^{12}] \delta_{11i2} - \\
& \frac{1}{432} g_1^2 g_2^2 \text{LF}_{3,1,-1}[\text{m}_1, \text{m}_q^{12}] \delta_{11i2} + \frac{1}{864} g_1^2 g_2^2 \text{LF}_{4,1,-2}[\text{m}_1, \text{m}_q^{12}] \delta_{11i2} + \\
& \frac{2}{9} g_1^2 s_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \text{LF}_{2,1,0}[\text{m}_1, \text{m}_d^p] - \frac{4}{9} g_1^2 s_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \text{LF}_{3,1,-1}[\text{m}_1, \text{m}_d^p] + \\
& \frac{2}{9} g_1^2 s_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \text{LF}_{4,1,-2}[\text{m}_1, \text{m}_d^p] - \frac{1}{96} g_1^4 \text{LF}_{2,1,0}[\text{m}_2, \text{m}_q^{11}] \delta_{11i2} - \\
& \frac{1}{96} g_1^4 \text{LF}_{2,2,-1}[\text{m}_2, \text{m}_q^{11}] \delta_{11i2} - \frac{5}{48} g_1^4 \text{LF}_{3,1,-1}[\text{m}_2, \text{m}_q^{11}] \delta_{11i2} + \frac{1}{32} g_1^4 \text{LF}_{4,1,-2}[\text{m}_2, \text{m}_q^{11}] \delta_{11i2} - \\
& \frac{1}{96} g_1^4 \text{LF}_{2,1,0}[\text{m}_2, \text{m}_q^{12}] \delta_{11i2} - \frac{1}{96} g_1^4 \text{LF}_{2,2,-1}[\text{m}_2, \text{m}_q^{12}] \delta_{11i2} - \frac{5}{48} g_1^4 \text{LF}_{3,1,-1}[\text{m}_2, \text{m}_q^{12}] \delta_{11i2} + \\
& \frac{1}{32} g_1^4 \text{LF}_{4,1,-2}[\text{m}_2, \text{m}_q^{12}] \delta_{11i2} - \frac{1}{3} g_1^2 \text{LF}_{2,1,0}[\text{m}_2, \tilde{\mu}] \delta_{11i2} - \frac{1}{3} g_1^2 \text{LF}_{2,2,-1}[\text{m}_2, \tilde{\mu}] \delta_{11i2} - \\
& \frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{2,2,0}[\text{m}_2, \tilde{\mu}] \delta_{11i2} + \frac{1}{6} g_2^4 \text{LF}_{3,1,-1}[\text{m}_2, \tilde{\mu}] \delta_{11i2} + \\
& \frac{1}{6} g_2^4 \text{LF}_{3,2,-2}[\text{m}_2, \tilde{\mu}] \delta_{11i2} + \frac{1}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,2,-1}[\text{m}_2, \tilde{\mu}] \delta_{11i2} + \\
& \frac{2}{3} g_3^2 c_\gamma^2 \overline{y}_d^{12p} y_d^{11p} \text{LF}_{2,1,0}[\text{m}_3, \text{m}_d^p] - \frac{4}{3} g_3^2 c_\gamma^2 \overline{y}_d^{12p} y_d^{11p} \text{LF}_{3,1,-1}[\text{m}_3, \text{m}_d^p] + \\
& \frac{2}{3} g_3^2 c_\gamma^2 \overline{y}_d^{12p} y_d^{11p} \text{LF}_{4,1,-2}[\text{m}_3, \text{m}_d^p] + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,1,0}[\text{m}_3, \text{m}_q^{11}] \delta_{11i2} + \\
& \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,2,-1}[\text{m}_3, \text{m}_q^{11}] \delta_{11i2} - \frac{1}{9} g_2^2 g_3^2 \text{LF}_{3,1,-1}[\text{m}_3, \text{m}_q^{11}] \delta_{11i2} + \\
& \frac{1}{18} g_2^2 g_3^2 \text{LF}_{4,1,-2}[\text{m}_3, \text{m}_q^{11}] \delta_{11i2} + \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,1,0}[\text{m}_3, \text{m}_q^{12}] \delta_{11i2} + \\
& \frac{1}{18} g_2^2 g_3^2 \text{LF}_{2,2,-1}[\text{m}_3, \text{m}_q^{12}] \delta_{11i2} - \frac{1}{9} g_2^2 g_3^2 \text{LF}_{3,1,-1}[\text{m}_3, \text{m}_q^{12}] \delta_{11i2} + \\
& \frac{1}{18} g_2^2 g_3^2 \text{LF}_{4,1,-2}[\text{m}_3, \text{m}_q^{12}] \delta_{11i2} + \frac{2}{3} g_3^2 s_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \text{LF}_{2,1,0}[\text{m}_3, \text{m}_d^p] - \\
& \frac{4}{3} g_3^2 s_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \text{LF}_{3,1,-1}[\text{m}_3, \text{m}_d^p] + \frac{2}{3} g_3^2 s_\gamma^2 \overline{y}_u^{12p} y_u^{11p} \text{LF}_{4,1,-2}[\text{m}_3, \text{m}_d^p] + \\
& \frac{1}{8} g_2^2 \left(c_\gamma \overline{a}_e^{pr} - s_\gamma \tilde{\mu} \overline{y}_e^{pr} \right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) \text{LF}_{4,1,-1}[\text{m}_q^p, \text{m}_e^r] \delta_{11i2} - \\
& \frac{1}{6} g_2^2 \left(c_\gamma \overline{a}_e^{pr} - s_\gamma \tilde{\mu} \overline{y}_e^{pr} \right) \left(c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr} \right) \text{LF}_{5,1,-2}[\text{m}_q^p, \text{m}_e^r] \delta_{11i2} + \\
& \frac{3}{8} g_2^2 \left(c_\gamma \overline{a}_d^{pr} - s_\gamma \tilde{\mu} \overline{y}_d^{pr} \right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) \text{LF}_{4,1,-1}[\text{m}_q^p, \text{m}_d^r] \delta_{11i2} - \\
& \frac{1}{2} g_2^2 \left(c_\gamma \overline{a}_d^{pr} - s_\gamma \tilde{\mu} \overline{y}_d^{pr} \right) \left(c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr} \right) \text{LF}_{5,1,-2}[\text{m}_q^p, \text{m}_d^r] \delta_{11i2} + \\
& \frac{1}{4} g_2^2 \left(s_\gamma \overline{a}_u^{pr} - \tilde{\mu} c_\gamma \overline{y}_u^{pr} \right) \left(s_\gamma a$$